MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 2X2A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Rectifier, high voltage, half wave, vacuum

Outline and dimensions: See figure 1
Cathode --- Coated unipotential
Mounting position: Any

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	Epp	ib	Io	Rl	Cl	i (surge)	Alt
Unit:	Vac	kv	kVac	ma	mAdc	Meg	uF	ma	ft
Maximum:	2.75	12.5	---	45	7.5	---	---	100	10,000
Minimum:	2.25	---	---	---	---	---	---	---	---
TEST CONDITIONS: (operation)	2.5	---	4.5	---	---	0.6	0.06	---	---

GENERAL:

First article sample inspection - Required (see note 5)

C denotes changes

2X2A
FSC 5960

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(C)

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1256	Electrode current (anode)	Eb = 200 Vdc; Ef = 2.5 V	0.65	II	Ib	30	65	mAdc
1353	Operation of rectifiers	See notes 1 and 2	0.65	II	Io	6.0	---	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1041	Shock	No voltages applied (see note 3)	15	S2	---	---	---	---
1301	Heater current	Ef = 2.5 V	---	---	If	1.55	1.95	A
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life	Group C (see note 4)	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours)							
---	Electrode current (anode) and Failure to rectify		---	---	Ib	20	---	mAdc

NOTES:

1. In a half-wave circuit adjust Zp such that a tube having Etd = 98 Vdc at 15 mAdc gives Io = 6.8 mAdc. Apply heater and anode voltage simultaneously to cold tube. After cathode has reach operating temperature key anode circuit three times.

2. This test shall be performed at the conclusion of the holding period.

3. The TUT shall be subjected to three blows of 300 G in each of the following positions; vertically, with base up; vertically, with base down; horizontally, with stem press perpendicular; horizontally, with stem press parallel. There shall be no stripped seals, loose bases, breakage, or objectionable displacement of parts. A stripped seal around the anode cap can be identified by a silvery line appearing at the seal.

4. During life-test operation, the values of R1 and C1 given in test conditions may be considered as approximate, and shall be adjusted initially to give Io = 7.5 mAdc with ib greater than 40 ma.

5. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this tube specification sheet. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Air Force - 99
DLA - ES

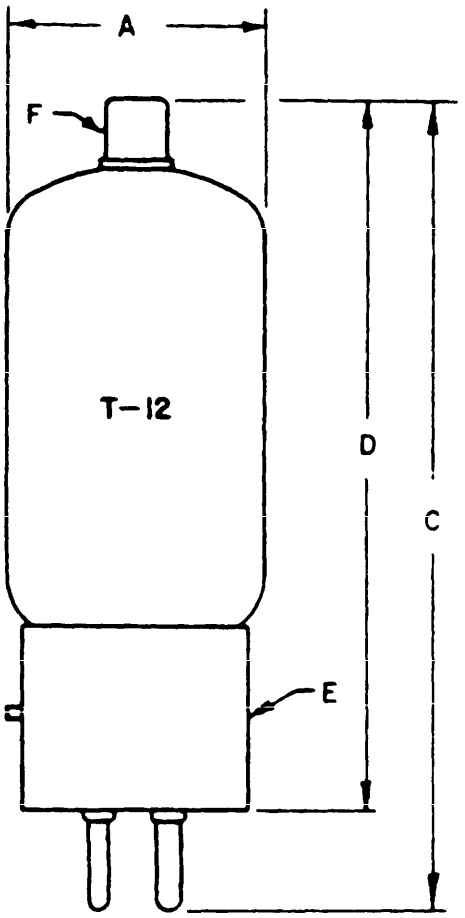
User activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3262)

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PIN CONNECTIONS	
Pin No.	Element
1	h
2	nc
3	nc
4	h,k
Cap	a

Dimensions				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
Quality conformance inspection, part 2				
A	---	1.5625	---	39.688
C	---	4.5312	---	115.092
D	3.6562	3.9062	92.867	99.217
Quality conformance inspection, part 3 (see note b)				
E	Base: A4-10 (see note a) (EIA)			
F	Cap: C1-1 or C1-34 (EIA)			

NOTES:
a. Base material shall be low-loss phenolic.
b. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Ⓒ FIGURE 1. Outline drawing.

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OMB Approval
No. 22-R255

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